

Total No. of Questions—8]

[Total No. of Printed Pages—3

Seat No.	
-------------	--

[5152]-165

S.E. (Computer Engg.) (First Semester) EXAMINATION, 2017
MICROPROCESSOR ARCHITECTURE
(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Assume suitable data, if necessary.

1. (a) Explain the test registers with their format. [3]
(b) Differentiate between Min mode and Max mode of 8086 processor [4]
(c) Explain the following pins of 8086 : [6]
(i) INTA#
(ii) HLDA
(iii) DT/R#

Or

2. (a) Specify size and function of IDTR. [3]
(b) Explain the hardware interrupt pins of 80386 processor. [4]
(c) Draw and explain the functional pin diagram of 80386 DX processor. [6]
3. (a) Draw and explain non-pipelined write cycle of 80386 DX. [5]
(b) Explain HOLD/HLDA signal. [4]

P.T.O.

- (c) Explain with example difference between SUB and SBB instruction. [3]

Or

4. (a) Explain the following instructions : [5]
- (i) XOR
 - (ii) TEST
 - (iii) WAIT
 - (iv) STD
 - (v) SHLD.
- (b) What do you mean by assembler directives ? Explain any *three* assembler directives used in 80386 programming. [4]
- (c) Explain the assembly language programming steps. [3]
5. (a) Explain in detail multiprocessor architecture. [6]
- (b) Give the parallel programming with diagram. [4]
- (c) Briefly explain hybrid multicore architecture. [3]

Or

6. (a) Draw and explain the block diagram of 64-bit architecture. [6]
- (b) Write a short note on multicore application design and implementation. [4]
- (c) Explain front side and black side bus. [3]

7. (a) Explain the IA-32 basic execution environment. [5]
(b) State the two sub-modes of IA-32e mode. [4]
(c) What are the advantages of hyper threading technology ? [3]

Or

8. (a) Explain in detail microarchitecture code name Nehalem. [5]
(b) List the features of SSE2 extensions in Pentium 4 and Intel Xeon processor. [4]
(c) Explain the features of Virtualization Technology. [3]